



Library & Archives Mobile App for Member Self-Services

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Database SystemsEngineering andDistributedBackendDevelopment(25CS1302E) | KLHBachupallyCampus, CSE | PBL 2026-27

Abstract

- A mobile self-service app for library members to browse, reserve and track books, fines and alerts.
- Offline access with background sync.

Problem Statement & Objectives

- Problem: Manual, slow library services.
- Objective 1: Anytime mobile access.
- Objective 2: Offline access and sync.
- Objective 3: Real-time alerts.
- Scope: Members, catalogue, reservations.

Existing vs Proposed System

Feature	Existing System	Proposed System
Access	Physical visits	Mobile app
Reservation	Manual	In-app
Alerts Offline	Manual reminders	FCM push alerts
use	Not available	Room cache
Limitation/Gap	Slow, less accessible	Self-service + sync

Feasibility Study

- Technical: Kotlin, Compose, Node.js, MySQL; all freely available.
- Economic: Open-source stack, free-tier Firebase.
- Operational: Simple mobile UI, easy to maintain.
- Schedule: Completed within the trimester.

Results & Testing

- CRUD and validation tested.
- API tests passed in Postman.

Conclusion

- Working self-service app.
- Meets all objectives.

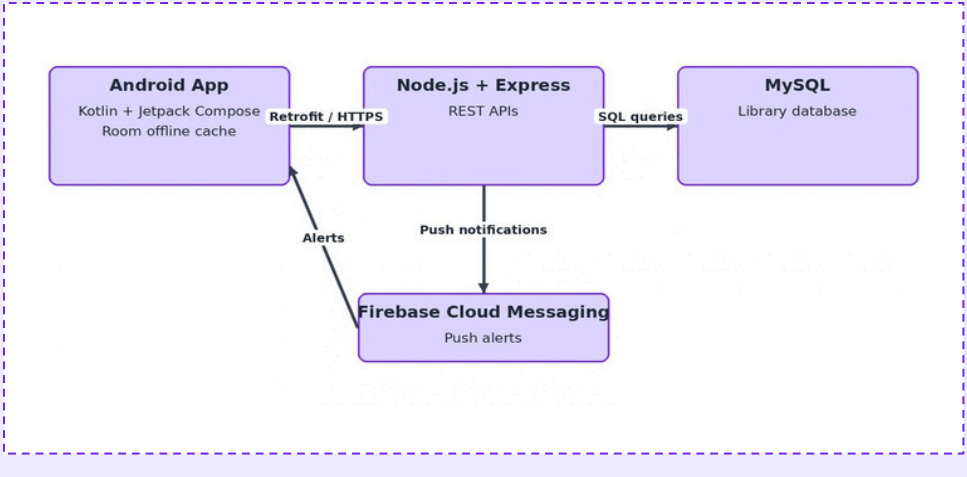
Future Scope

- Barcode/ QR scanning
- Online fine payment

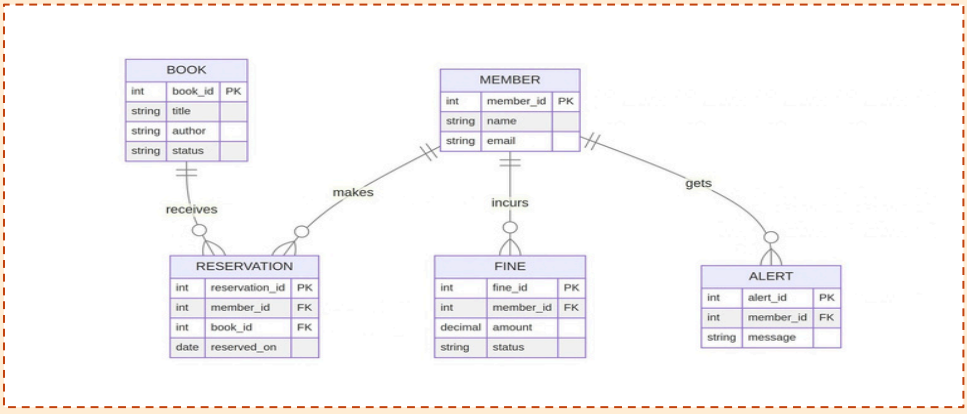
Tools & Technologies

- Front-End: Android (Kotlin) + Jetpack Compose
- Back-End: Node.js + Express.js REST APIs, Retrofit
- Database: MySQL (server), Room (offline cache)
- Tools: Firebase Cloud Messaging, Postman, GitHub, Android Studio

System Architecture



ER Diagram & Database Design



Implementation Screenshots

